

AI Hospital Appointment Orchestrator

1. Product Vision

The AI Hospital Appointment Orchestrator is designed for a **single hospital**. The hospital manages its doctors, departments, schedules, and appointments through an admin dashboard. Patients interact with an AI assistant using voice or text to book, reschedule, cancel, or check appointments. The AI understands symptoms, recommends the appropriate department and doctor, checks availability, books the appointment, and sends confirmation notifications.

2. User Roles

- **Patient** – Book, reschedule, cancel, and view appointments using AI.
 - **Hospital Admin** – Manage doctors, departments, schedules, appointments, and hospital settings.
 - **Doctor** – View daily schedule and upcoming appointments.
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3. Hospital Data

Dummy data :

Here's a realistic **dummy hospital dataset** that you can include in your SRS. It represents a medium-sized private hospital.

Hospital Information

Hospital Name: Sunrise Multispeciality Hospital

Address: 123 Health Avenue, Ahmedabad, Gujarat 380015

Phone: +91 79 4012 3456

Email: contact@sunrisehospital.com

Website: www.sunrisehospital.com

Working Days: Monday – Saturday

Working Hours: 8:00 AM – 8:00 PM

Emergency Department: 24 × 7

Supported Languages: English, Hindi, Gujarati

Departments

Department	Floor	Description
General Medicine	1	Fever, cold, infections, general illness
Cardiology	2	Heart-related diseases
Orthopedics	2	Bones, joints, fractures
Neurology	3	Brain and nervous system
Pediatrics	3	Child healthcare
Dermatology	4	Skin, hair, nails
ENT	4	Ear, Nose & Throat

Gynecology	5	Women's healthcare
Ophthalmology	5	Eye care
Dentistry	Ground	Dental care

Doctors

Doctor	Department	Experience	Languages	Consultation Fee
Dr. Rahul Shah	General Medicine	12 Years	English, Hindi, Gujarati	\$15
Dr. Priya Mehta	Cardiology	15 Years	English, Hindi	\$30
Dr. Amit Patel	Orthopedics	10 Years	English, Gujarati	\$20
Dr. Neha Desai	Pediatrics	9 Years	English, Hindi, Gujarati	\$18
Dr. Kunal Joshi	Neurology	14 Years	English, Hindi	\$35

Dr. Riya Shah	Dermatology	8 Years	English, Gujarati	\$18
Dr. Harsh Trivedi	ENT	11 Years	English, Hindi	\$18
Dr. Anjali Patel	Gynecology	13 Years	English, Hindi, Gujarati	\$25
Dr. Vivek Modi	Ophthalmology	9 Years	English, Hindi	\$18
Dr. Chirag Bhatt	Dentistry	10 Years	English, Gujarati	\$15

Doctor Working Schedule

Doctor	Days	Time
Dr. Rahul Shah	Mon–Sat	9:00 AM – 4:00 PM
Dr. Priya Mehta	Mon–Fri	10:00 AM – 5:00 PM
Dr. Amit Patel	Mon–Sat	9:00 AM – 3:00 PM

Dr. Neha Desai	Mon–Sat	10:00 AM – 6:00 PM
Dr. Kunal Joshi	Tue–Sat	11:00 AM – 5:00 PM
Dr. Riya Shah	Mon–Fri	9:00 AM – 2:00 PM
Dr. Harsh Trivedi	Mon–Sat	1:00 PM – 7:00 PM
Dr. Anjali Patel	Mon–Sat	10:00 AM – 4:00 PM
Dr. Vivek Modi	Mon–Fri	9:00 AM – 1:00 PM
Dr. Chirag Bhatt	Mon–Sat	9:00 AM – 6:00 PM

Appointment Rules

- Appointment Duration: **30 Minutes**
- Buffer Between Appointments: **5 Minutes**
- Maximum Patients Per Doctor Per Day: **16**
- Online Booking Window: **Up to 30 Days in Advance**
- Same-Day Booking: **Allowed if slots are available**
- Cancellation Deadline: **At least 2 hours before the appointment**
- Rescheduling Limit: **Maximum 2 times per appointment**

Hospital Holidays

- Republic Day (26 January)
- Holi
- Independence Day (15 August)
- Gandhi Jayanti (2 October)
- Diwali (2 Days)
- Christmas (25 December)

Note: Emergency Department remains operational on all holidays.

Doctor Languages

Lang uage	Available Doctors
Englis h	All Doctors
Hindi	Dr. Rahul Shah, Dr. Priya Mehta, Dr. Neha Desai, Dr. Kunal Joshi, Dr. Harsh Trivedi, Dr. Anjali Patel, Dr. Vivek Modi
Gujar ati	Dr. Rahul Shah, Dr. Amit Patel, Dr. Neha Desai, Dr. Riya Shah, Dr. Anjali Patel, Dr. Chirag Bhatt

Hospital Contact Information

Reception: +91 79 4012 3456

Emergency: +91 79 4012 3999

Email: contact@sunrisehospital.com

Appointment Email: appointments@sunrisehospital.com

Address: 123 Health Avenue, Ahmedabad, Gujarat 380015

Support Hours: Monday–Saturday, 8:00 AM – 8:00 PM

This dataset is realistic enough for development, demonstrations, testing, and AI-agent implementation. It provides sufficient data for doctor recommendations, schedule optimization, language matching, appointment booking, and notification workflows without making the project unnecessarily complex.

4. Complete User Flow

Patient opens website



Starts AI chat (voice/text)



AI understands symptoms



Collects missing information



Detects emergency if required



Recommends doctor



Finds available slots



Patient confirms



Appointment booked



Confirmation email sent

5. Frontend Pages

- Landing Page
 - Login/Register
 - AI Chat
 - Appointment Confirmation
 - Appointment History
 - Admin Dashboard
 - Doctor Dashboard
 - Profile
-

6. Business Rules

- One hospital only.
 - One appointment per doctor per time slot.
 - Doctors can only be booked during working hours.
 - Hospital holidays cannot be booked.
 - Emergency cases are not automatically booked.
 - Duplicate appointments should be prevented.
-

Software Requirements & Technical Design Document (SRS)

Version: 1.0

Technology: LangGraph + FastAPI + React + PostgreSQL

1. Project Overview

1.1 Project Name

AI Hospital Appointment Orchestrator

1.2 Project Description

The AI Hospital Appointment Orchestrator is an intelligent appointment booking platform powered by LangGraph and Large Language Models.

Instead of filling multiple forms, a patient can simply speak or type naturally.

Example:

I have had chest pain since yesterday. I want an appointment tomorrow morning with a cardiologist.

The AI understands the request, extracts symptoms, detects emergencies, finds the correct department, checks doctor schedules, books the appointment, and notifies the patient.

The system uses multiple AI agents coordinated through LangGraph rather than a single chatbot.

2. Problem Statement

Current hospital appointment systems require users to:

- Search departments manually
- Search doctors manually
- Choose appointment slots manually
- Fill lengthy forms
- Repeat information multiple times

Hospitals also struggle with:

- Wrong department selection
- Missed appointments
- Schedule conflicts

- Poor patient experience
- Emergency cases being booked instead of redirected

The project aims to automate these processes using AI agents.

3. Objectives

Primary objectives

- Build a production-like AI application using LangGraph
- Demonstrate multi-agent orchestration
- Reduce manual appointment booking
- Detect emergency situations automatically
- Recommend appropriate doctors
- Optimize appointment scheduling

Secondary objectives

- Voice interaction
 - Multi-language support
 - Conversation memory
 - Modular AI architecture
 - Easy extensibility
-

4. Project Scope

Included

- Appointment booking
- Appointment rescheduling
- Appointment cancellation
- Voice input
- Doctor recommendation
- Emergency detection
- Schedule optimization
- Email notification
- Doctor dashboard

Not Included

- Online payment
 - Video consultation
 - Prescription generation
 - Medical diagnosis
 - Electronic Health Records (EHR)
 - Insurance claim processing
-

5. Technology Stack

Frontend

- React
- Tailwind CSS
- Axios

Backend

- FastAPI
- Python

AI

- LangGraph
- LangChain
- OpenRouter (GPT/Claude/Gemini)

Database

- PostgreSQL

ORM

- SQLAlchemy

Authentication

- JWT

Email

- Gmail SMTP

Voice

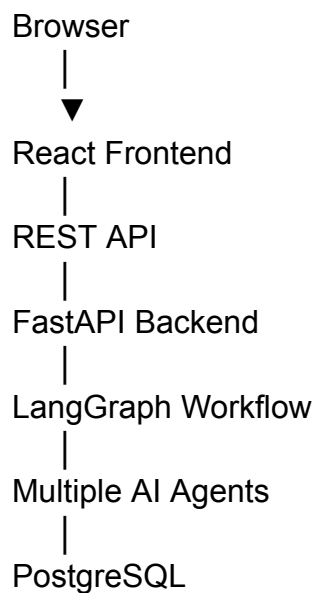
Speech Input

- Browser Speech Recognition API

Speech Output

- Browser Speech Synthesis API
-

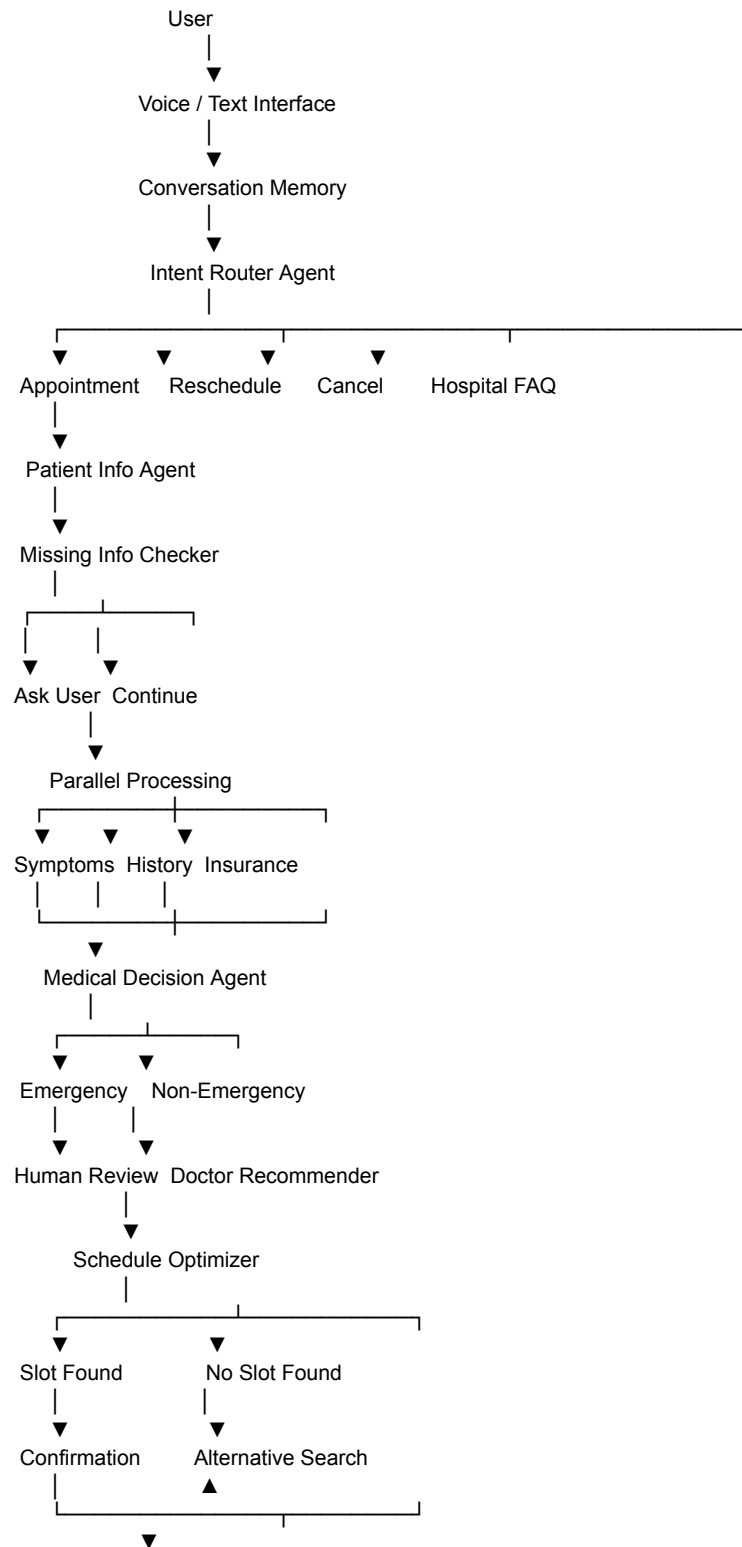
6. High-Level System Architecture

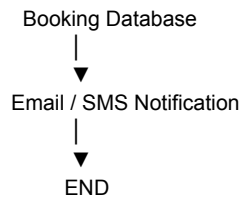


The frontend never communicates directly with AI.

Everything passes through FastAPI.

7. LangGraph Architecture





8. Agent Specifications

8.1 Input Agent

Purpose

Receive user text or speech.

Input

Raw sentence

Output

Clean text

8.2 Intent Agent

Responsibilities

Identify user intention.

Possible intents

- Book appointment
- Cancel appointment
- Reschedule
- Check appointment

Output

intent=BOOK

8.3 Patient Details Agent

Extract

- Name
- Age
- Gender
- Preferred language
- Preferred time

If missing

Return

Need more information

8.4 Missing Information Agent

Checks

Age?

Symptoms?

Preferred date?

Preferred time?

Department?

If information is missing

Loop back to user.

8.5 Symptom Extraction Agent

Input

Natural language

Output

Symptoms

Duration

Severity

Department

Example

Symptoms

Fever

Headache

Department

General Physician

8.6 Evaluator Agent

Purpose

Validate extraction quality.

If confidence

<80%

Ask additional questions.

Else

Continue.

8.7 Emergency Agent

Detect

- Heart attack
- Stroke
- Heavy bleeding
- Breathing problem

If emergency

Skip booking.

Recommend emergency room.

8.8 Doctor Recommendation Agent

Reads database.

Filters

- Department
- Language
- Availability

Returns ranked doctors.

8.9 Schedule Agent

Checks

Doctor schedule

Hospital holidays

Existing appointments

Finds best slot.

8.10 Booking Agent

Creates appointment.

Updates schedule.

Stores booking.

8.11 Notification Agent

Sends

- Email
 - Confirmation
 - Appointment summary
-

9. Entity & Database Design

User

- id
 - name
 - email
 - phone
 - age
 - gender
 - preferred_language
-

Doctor

- id
 - name
 - specialization
 - department
 - experience
 - languages
 - hospital
-

Schedule

- id
 - doctor_id
 - date
 - start_time
 - end_time
 - status
-

Appointment

- id
 - patient_id
 - doctor_id
 - schedule_id
 - symptoms
 - booking_status
 - created_at
-

Conversation

- id
- user_id
- messages
- current_state
- updated_at

10. API Specifications

POST

/api/chat

Receive user message.

POST

/api/appointment/book

Create booking.

POST

/api/appointment/cancel

Cancel appointment.

POST

/api/appointment/reschedule

Update booking.

GET

/api/doctors

Return doctor list.

GET

/api/schedule/{doctorId}

Return doctor schedule.

GET

/api/appointments/{userId}

Return appointment history.

11. State Management

LangGraph State

user_input

intent

language

patient

symptoms

department

priority

doctor

available_slots

selected_slot

booking_status

messages

State is shared across all agents.

12. User Flow

Booking

User



Voice



AI



Extract Symptoms



Emergency?



No



Doctor



Schedule



Confirmation



Booking

Emergency

Symptoms



Emergency



Hospital Alert



Stop Booking

13. Functional Requirements

The system shall

- Accept voice input
- Accept text input
- Detect language
- Understand symptoms
- Recommend department
- Detect emergencies
- Suggest doctors
- Search schedules
- Book appointments
- Cancel appointments
- Reschedule appointments

- Send email confirmation
 - Maintain conversation memory
-

14. Non-Functional Requirements

Performance

- Response time under 3 seconds for standard requests

Reliability

- Booking transactions should not create duplicate appointments

Security

- JWT authentication
- Password hashing
- SQL injection protection
- Input validation

Scalability

- Modular agent architecture
- Stateless API layer
- Horizontal backend scaling

Maintainability

- Each agent has a single responsibility
 - Independent testing for each node
-

15. Folder Structure

agents/

Contains every LangGraph node

graph/

Workflow definition

services/

Business logic

repositories/

Database access

schemas/

Pydantic models

prompts/

Prompt templates

utils/

Shared helpers

16. Development Guidelines

- One responsibility per agent
 - Business logic must not be placed in API routes
 - Agents should communicate only through LangGraph state
 - Keep prompts version-controlled
 - Validate every external input
 - Use transactions for booking operations
 - Follow REST API conventions
 - Write unit tests for each agent
-

17. Testing Strategy

Unit Testing

- Test each agent independently

Integration Testing

- Complete LangGraph workflow

API Testing

- FastAPI endpoints

UI Testing

- React components

Edge Cases

- Missing age
 - Invalid symptoms
 - Emergency case
 - No doctors available
 - No available slot
 - Duplicate booking
 - Network failure
 - Invalid authentication
-

18. Deployment

Frontend

- Vercel

Backend

- Render or Railway

Database

- Neon PostgreSQL

Environment Variables

OPENROUTER_API_KEY

DATABASE_URL

JWT_SECRET

EMAIL_USER

EMAIL_PASSWORD

19. Future Enhancements

- Online payment
- Video consultation
- AI medical report summarization
- OCR for prescriptions
- Wearable integration
- WhatsApp notifications
- SMS reminders
- Insurance verification
- Multi-hospital support
- Doctor AI assistant
- Analytics dashboard
- Admin portal
- Mobile application

20. Appendix

Assumptions

- Doctors maintain accurate schedules.

- Patients provide truthful symptom information.
- Emergency detection is advisory and does not replace medical diagnosis.

Risks

- LLM misclassification of symptoms
- Schedule conflicts due to concurrent bookings
- API outages
- Voice recognition inaccuracies

Success Criteria

- Patients can book appointments in under 2 minutes.
 - Emergency cases are routed correctly.
 - No double-booking of doctor slots.
 - Modular agent design supports future expansion with minimal changes.
-

Interview Highlights

This project demonstrates:

- LangGraph multi-agent orchestration
- Conditional routing
- State management
- Conversation memory
- Loop-based information collection
- Human-like conversational booking
- PostgreSQL-backed scheduling
- FastAPI backend architecture
- React frontend integration
- Production-oriented modular design

Search :
exact DB schema

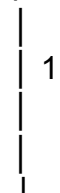
users

id

name

email

phone



1



N

appointments

id

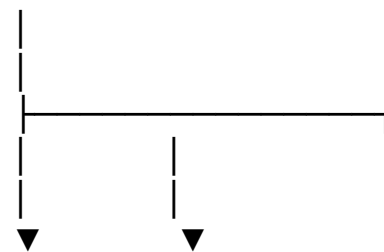
patient_id

doctor_id

schedule_id

symptoms

status



doctors

doctor_schedules

id

department

name

fee



id

doctor_id

date

time

departments

id

name

Department

1

↓

Many Doctors

Doctor

1

↓

Many Schedules

Doctor

1

↓

Many Appointments

Patient

1

↓

Many Appointments

Conversation

1

↓

Many Messages

graph state
node interfaces

11. LangGraph State

11.1 Purpose

The LangGraph State acts as the **single source of truth** during workflow execution. Every agent reads from and writes to this shared state instead of communicating directly with each other.

This enables:

- Stateful conversations
- Conditional routing
- Parallel execution
- Retry loops
- Human approval
- Workflow resumption
- Checkpointing

11.2 Graph State Schema

HospitalState

session_id: str

user_id: UUID

messages: List[Message]

language: str

intent: str

patient: {

 name

 age

 gender

 phone

}

symptoms: {

 symptoms[]

 duration

 severity

 body_part

}

department: str

priority: LOW | MEDIUM | HIGH | EMERGENCY

doctor_candidates: []

selected_doctor

available_slots: []

selected_slot

appointment_status

booking_id

requires_human_approval

retry_count

errors: []

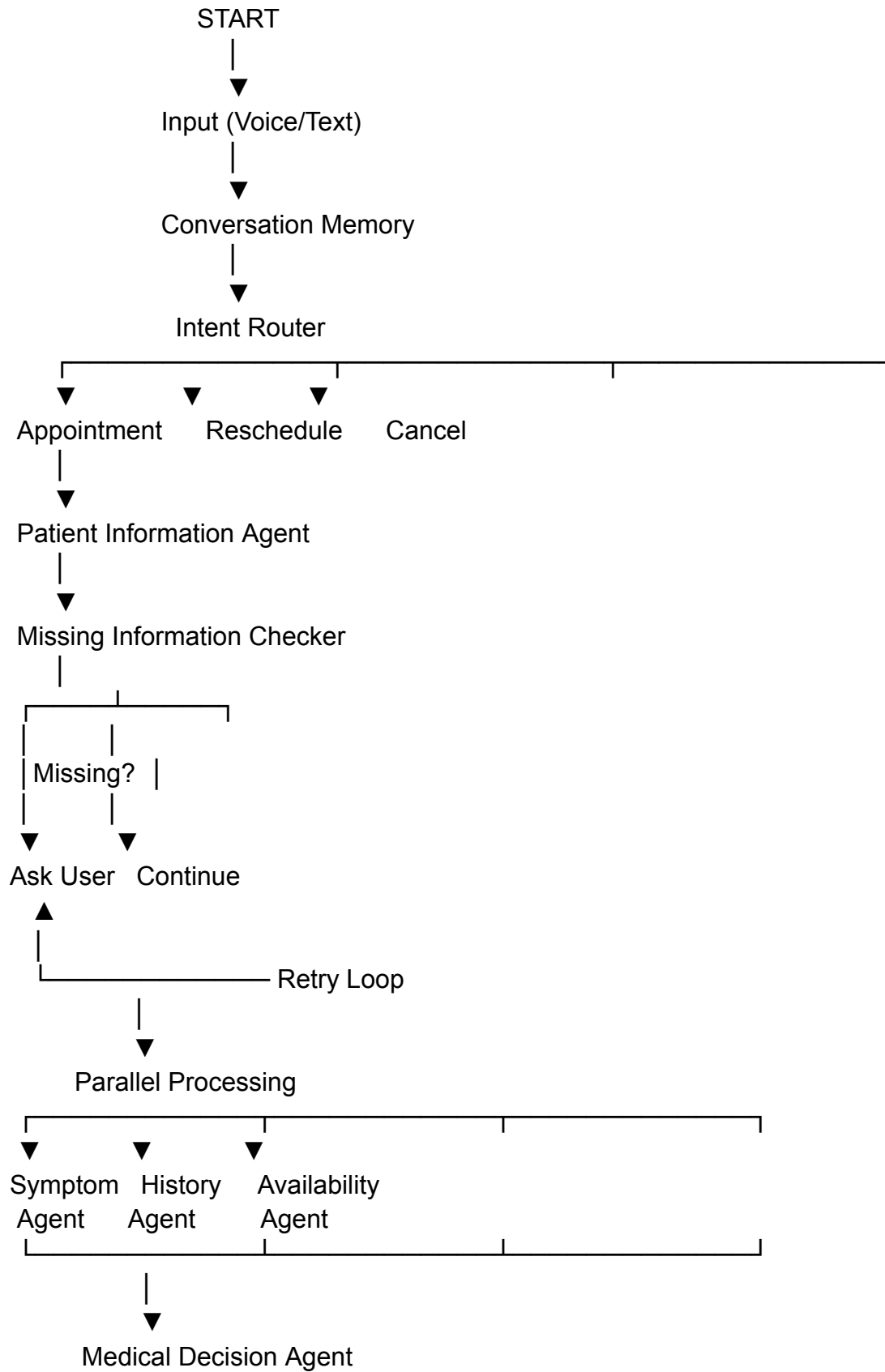
metadata

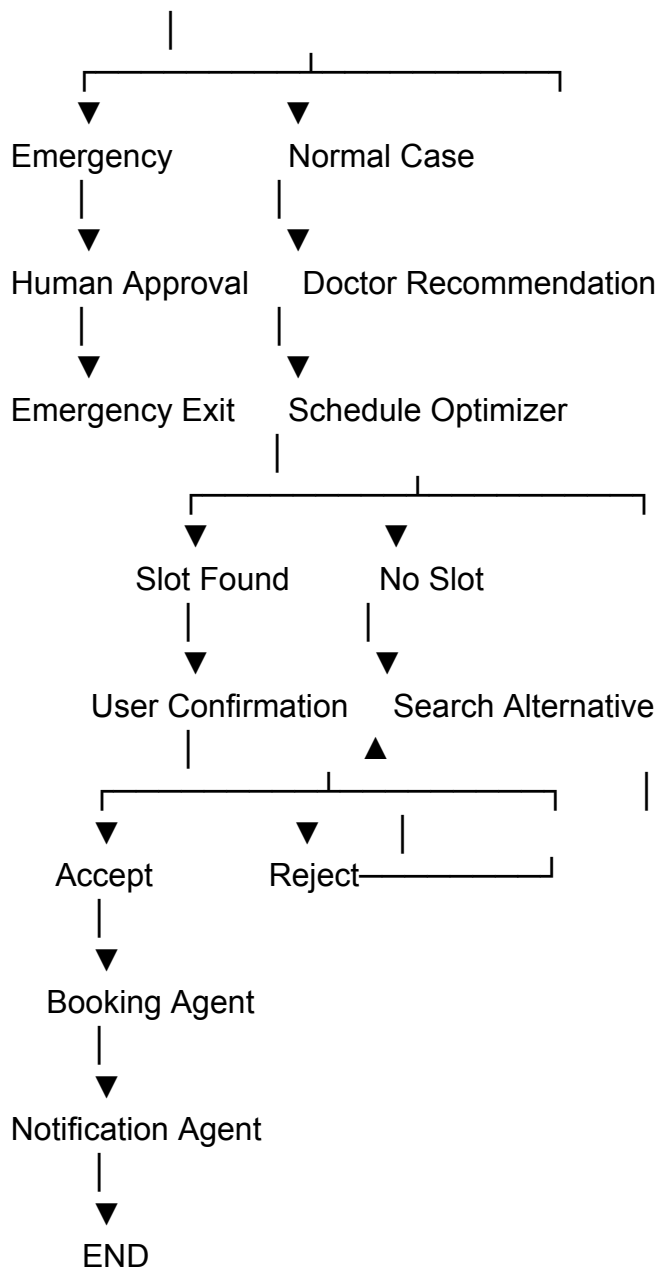
11.3 State Ownership

State	Updated By
intent	Intent Agent
patient	Patient Info Agent
symptoms	Symptom Agent
priority	Emergency Agent
doctor_candidates	Doctor Agent
available_slots	Schedule Agent
selected_slot	Confirmation Agent
booking_status	Booking Agent
errors	Any Agent

12. LangGraph Workflow

High-Level Graph

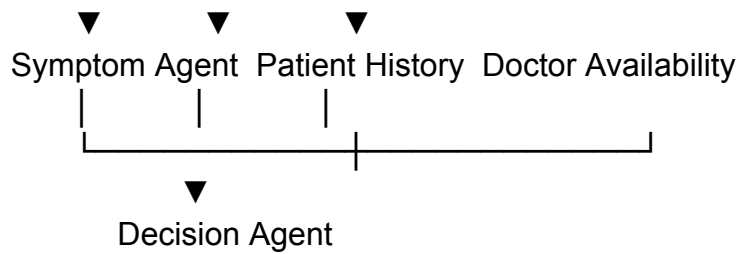




13. Parallel Execution

The following agents execute simultaneously to reduce latency.





Advantages

- Lower response time
 - Independent execution
 - Better scalability
-

14. Decision Nodes

Intent Router

Input

"I want appointment tomorrow."

Output

BOOK_APPOINTMENT

Possible Routes

- Appointment
 - Cancel
 - Reschedule
 - Check Status
-

Missing Information

Decision

Age Available?

YES



Continue

NO



Ask User

Emergency Detection

Decision

Emergency?

YES



Human Approval



Emergency Exit

NO



Doctor Recommendation

Slot Availability

Decision

Slot Available?

YES

↓

Book

NO

↓

Alternative Search

User Confirmation

Decision

Accept Slot?

YES

↓

Booking

NO

↓

Search Again

15. Retry Loops

Missing Information Loop

Extract Information



Required Fields Missing



Ask User



Update State



Revalidate



Continue

Doctor Search Loop

Search Doctor



Unavailable



Expand Search



Nearby Doctors



Different Time



Next Day



Return Results

Slot Confirmation Loop

Offer Slot



Rejected



Find Next Slot



Offer Again

Maximum retries: **3**

16. Human-in-the-Loop

Human intervention is required when AI confidence is low or the request has safety implications.

Triggers

- Emergency detected
- Low confidence (<80%)
- Invalid patient information
- Manual override

Workflow

AI Decision

↓

Pause Graph

↓

Hospital Staff Review

↓

Approve / Reject

↓

Resume Graph

17. Node Interfaces

Input Agent

Input

Voice/Text

Output

clean_text
language

Intent Agent

Input

clean_text

Output

intent

Patient Information Agent

Input

clean_text

Output

patient
missing_fields

Symptom Agent

Input

clean_text
patient

Output

symptoms

department
severity

History Agent

Input

patient_id

Output

previous_doctors
previous_departments
medical_history

Availability Agent

Input

department
preferred_time

Output

doctor_candidates

Medical Decision Agent

Input

symptoms
history

Output

priority
confidence

Emergency Agent

Input

priority

Output

emergency=True/False

Doctor Recommendation Agent

Input

department
history
availability

Output

selected_doctor

Schedule Optimizer

Input

doctor
preferred_date
preferred_time

Output

available_slots

Confirmation Agent

Input

available_slots

Output

selected_slot

Booking Agent

Input

doctor

slot

patient

Output

booking_id

booking_status

Notification Agent

Input

booking

Output

email_sent

sms_sent

18. Graph Characteristics

Feature	Implementation
State Management	Shared <code>HospitalState</code>
Conditional Routing	Intent, Emergency, Slot Availability
Parallel Execution	Symptom, History, Availability
Retry Loops	Missing information, Slot search
Human Approval	Emergency & Low Confidence
Interrupt/Resume	LangGraph Checkpoint
Memory	Conversation + Booking Context
Error Recovery	Retry + Fallback Path
Extensibility	Add new agents without changing existing nodes

This graph is **significantly more representative of LangGraph** than a linear sequence because it includes:

- **Parallel branches** (independent agents running concurrently)
- **Conditional routing** (intent, emergency, slot availability)
- **Retry loops** (collecting missing information, finding alternative slots)
- **Human-in-the-loop** checkpoints
- **Shared graph state** accessed by all agents
- **Checkpoint/resume** capability
- **Modular node interfaces** that can be extended without redesigning the workflow

edge cases

19. Node-wise Edge Cases & Handling

Node	Edge Case	Detection	Action	Next Node
Input Agent	Empty input	Input length = 0	Ask user to speak/type again	Input Agent
	Voice recognition failed	STT returns null	Retry speech recognition (max 2)	Input Agent
	Unsupported language	Language not detected	Default to English & ask confirmation	Intent Agent
	Very long input	>2000 characters	Summarize input before processing	Intent Agent
	Offensive/irrelevant text	Intent confidence < threshold	Ask user to rephrase	Input Agent

Intent Agent

Edge Case	Detection	Action	Next Node
Multiple intents ("Book and cancel")	Multiple intent score > threshold	Ask user to choose one task	Intent Agent
Unknown intent	Confidence < 70%	Ask clarification	Intent Agent
Greeting only	Greeting detected	Respond and wait	END
Emergency request mixed with booking	Emergency keywords present	Route to Emergency Agent	Emergency Agent

Patient Information Agent

Edge Case	Detection	Action	Next Node
Missing age	Age null	Ask age	Patient Info Agent
Missing gender	Gender null	Ask gender (optional)	Continue

Invalid age	Age <0 or >120	Ask again	Patient Info Agent
Missing patient name	Name null	Ask name	Patient Info Agent
Returning patient	Existing account found	Auto-fill profile	Symptom Agent

Missing Information Agent

Edge Case	Detection	Action	Next Node
Required field missing	Mandatory fields null	Ask only missing fields	Patient Info Agent
User refuses answer	Retry count >2	Continue with defaults if safe	Symptom Agent
User changes previous answer	Updated state	Overwrite state	Validation

Symptom Agent

Edge Case	Detection	Action	Next Node
Very vague symptom ("Not feeling good")	Low confidence	Ask follow-up questions	Symptom Agent
Contradictory symptoms	Medical inconsistency	Ask clarification	Symptom Agent
Multiple diseases possible	Confidence spread	Forward all candidates	Medical Decision
Non-medical request	Intent mismatch	Return to Intent Router	Intent Router
LLM extraction failed	JSON parse error	Retry prompt once	Symptom Agent

History Agent

Edge Case	Detection	Action	Next Node
First-time patient	No history	Continue normally	Medical Decision
Previous emergency visits	History flag	Increase priority score	Medical Decision
Database unavailable	DB timeout	Continue without history	Medical Decision

Availability Agent

Edge Case	Detection	Action	Next Node
No doctors in department	Empty result	Expand nearby hospitals	Doctor Recommendation
Doctor on leave	Schedule unavailable	Remove from candidates	Doctor Recommendation
Database timeout	Query timeout	Retry once	Availability Agent

Medical Decision Agent

Edge Case	Detection	Action	Next Node
Low confidence (<80%)	Confidence score	Human review or ask questions	Human Approval
High-risk symptoms	Emergency score	Route immediately	Emergency Agent
Conflicting data	Validation failed	Ask clarification	Symptom Agent

Emergency Agent

Edge Case	Detection	Action	Next Node
Chest pain + breathing issue	Rule engine/LLM	Stop booking	Emergency Exit
Stroke symptoms	Medical rule	Display emergency message	END
User insists on booking	Override attempt	Require human approval	Human Approval
False emergency	Human rejects	Resume booking	Doctor Recommendation

Human Approval Node

Edge Case	Detection	Action	Next Node
Staff approves	Approval received	Resume graph	Doctor Recommendation
Staff rejects	Rejected	Stop workflow	END
No response within timeout	Timeout	Escalate to supervisor	Human Approval

Doctor Recommendation Agent

Edge Case	Detection	Action	Next Node
Multiple equally suitable doctors	Same ranking	Return top 3 options	User Confirmation
Preferred doctor unavailable	Doctor unavailable	Suggest alternatives	Schedule Agent
No matching specialist	Empty list	Suggest General Physician	Schedule Agent

Schedule Agent

Edge Case	Detection	Action	Next Node
Requested slot unavailable	Slot occupied	Search nearest slot	Schedule Agent
Hospital holiday	Holiday calendar	Suggest another date	User Confirmation
Concurrent booking	Optimistic lock failed	Refresh availability	Schedule Agent
Doctor reached daily limit	Capacity reached	Next available day	Schedule Agent

Confirmation Agent

Edge Case	Detection	Action	Next Node
User rejects slot	Rejected	Search alternative	Schedule Agent
User changes date	New request	Restart scheduling	Schedule Agent
User changes symptoms	Medical info changed	Re-run medical decision	Symptom Agent

Booking Agent

Edge Case	Detection	Action	Next Node
Duplicate booking	Existing appointment	Ask for confirmation	Confirmation Agent
Database transaction failed	SQL error	Rollback transaction	Booking Agent
Slot booked simultaneously	Constraint violation	Refresh slots	Schedule Agent

Doctor becomes unavailable	Status changed	Find another doctor	Doctor Recommendation
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Notification Agent

Edge Case	Detection	Action	Next Node
Email failed	SMTP failure	Retry (3 attempts)	Notification Agent
SMS failed	Provider failure	Send email only	END
Notification duplicated	Existing log	Ignore duplicate	END

20. Global Graph Edge Cases

Scenario	Graph Behavior
User disconnects mid-conversation	Save checkpoint and resume later
LLM API unavailable	Retry → fallback model → notify user
Database unavailable	Retry with exponential backoff → maintenance message
Network timeout	Retry affected node only
User inactive for 10 minutes	Pause workflow and save state
Conversation exceeds token limit	Summarize history and continue
Multiple browser tabs	Lock session to prevent conflicting state updates
User starts a new booking while one is active	Ask whether to continue existing workflow or start a new session
Unexpected node exception	Log error, save state, route to Recovery Node

Recovery & Retry Policy

Failure Type	Retry Count	Strategy
LLM API failure	2	Retry → fallback model
Database timeout	3	Exponential backoff
Email failure	3	Retry asynchronously
Voice recognition	2	Prompt user to repeat
JSON parsing error	1	Re-prompt LLM for valid JSON
Human approval timeout	1	Escalate to supervisor
Slot conflict	Unlimited within request	Re-run scheduling until alternatives are exhausted